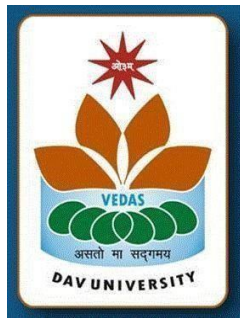


SIX Weeks Industrial Training Project Report on

ONLINE MOVIE BOOKING

Submitted in the partial fulfilment of the requirement for the award of degree of

Bachelor of Technology
in
Computer Science and Engineering
Batch (2022-2026)



Submitted to:
Ms.Bindu
(Assistant Professor)

Submitted by:
Amit Kumar
(12200324)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DAV UNIVERSITY

JALANDHAR- PUNJAB 144012

ABSTRACT

The purpose of this project is to develop an online movie ticket booking app. It is a system that enables customer to book movie ticket online at any anytime at any place. The Ticket Reservation System is an Internet based application that can be accessed throughout the Net and can be accessed by anyone who has a net connection. This application will reserve the tickets. This online ticket reservation system provides a website for a cinema hall where any user of internet can access it. User is required to login to the system and needs a credit card for booking the tickets. Tickets can be collected at the counter and Watching movies with family and friends in theatres is one of the best medium of entertainment after having a hectic schedule. The reason to develop the system is to help customer book tickets online. These helps in issues like peak hour-long queue issues. Therefore this system enhances the speed and standardization.

ACKNOWLEDGEMENT

If words are considered as a symbol of approval and token of appreciation then let the words play the heralding role expressing my gratitude.

First of all I thank to that Gracie God who blessed me with all kinds of facilities that had been provided to me for completion of my report.

I acknowledge my deepest sense of gratitude and sincere feeling of indebted divine my advisor Mr. Vipin Kumar under whose guidance and through their sustained efforts and encouraging attitude I was able to complete my project. It would have been difficult to achieve the results in such a short span of time.

I want to express my sincere gratitude to all the staff members of Netmax Technologies limited for spending their precious time and sharing the valuable information with me and in helping my project to be a success.

Specially Thanks to Mr. Manoj Kumar Vice Chancellor of D.A.V. University(Jalandhar) and faculty **Miss Bindu**(Assistant Professor) for encouragement to me. Words are not sufficient to register my sincere regards to my loving family & my friends for their cooperation & unabated inspiration that really keep me going.

Amit kumar

Declaration

I, Amit Kumar , hereby declare that the work which is being presented in this training titled “online movie Booking” by me, in partial fulfilment of the requirements for the award of Bachelor of Technology (B.Tech) Degree in “Computer Science and Artificial Intelligence” is an authentic record of my own work carried out under the guidance of **Ms. Bindu** (Assistant Professor). To the best of my knowledge, the matter embodied in this report has not been submitted to any other University/ Institute for the award of any degree.

Student Name: - Amit kumar Signature:

-

CERTIFICATE



CERTIFICATE OF COMPLETION

THIS IS TO CERTIFY THAT

Anirudh Kumar

has successfully completed training in

Web Development using ReactJS
JUN 2025 to JUL 2025

DIRECTOR

[Signature]

Date: 19-07-2025



ISO 9001:2008 certified company

FACULTY

Ref.No: NM/25/349

Table of Content

<i>Sr.no.</i>	<i>Contents</i>	<i>Page no.</i>
1	Chapter 1	
1.1	Introduction	1
1.2	Objectives	2
1.3	Organization	2-4
2	Chapter2 Literature Survey	5
3	Chapter3 objectives	6-8
4	Chapter4 Step to achieve objective	
5	Chapter 5	
5.1	Algorithm	11
5.2	Project Structure	12-15
6	Chapter6 Code Structure	16-17
7	Chapter 7	
7.1	Conclusion	18
7.2	Future Scope	19-20

List of Figures

Sr.	Figures	Page no.
1	Front design	13
2	Folder structure	13
3	Admin dashboard	14
4	Book detail	14
5	Output screen	15

COMPANY PROFILE



Established in 2001, NETMAX TECHNOLOGIES specializes in Network Support, Network Training, Software Training, and Embedded Systems.

In the education sector, we have strategic partnerships with Pearson VUE and Prometric. We are authorized testing partners for REDHAT and CISCO. Additionally, as a NOVELL EDUCATION PARTNER, we offer courses in NOVELL and SUSE LINUX. We also provide training in CADENCE-based design tools.

NETMAX TECHNOLOGIES offers Technical Research & Development support and consultancy services to various electronics companies. Our R&D clients in embedded systems include:

- Recorders and Medicare Ltd, Chandigarh
- TELEBOX India Ltd.
- Lotus Machines Pvt. Ltd., Chandigarh
- Impearl Electronics Pvt. Ltd., Chandigarh
- KANTA Electrical Ltd., Mohali

Our network services clients include:

- CEDTI, Mohali
- Premier ISP, Chandigarh
- Innovative Solutions, Chandigarh
- Emmtel ISP, Chandigarh
- NIPER, Mohali
- Navik Technologies, Chandigarh
- Software Technology Parks India, Mohali
- Glide Internet Services
- Rana Group
- IDS
- HFCL Infotel Ltd.
- Targus Technologies Pvt Ltd
- STPI, Mohali
- BBMB
- The Tribune

OUR TEAM

We are a dedicated team of certified technical professionals based in Chandigarh and Punjab, specializing in delivering comprehensive solutions. Our skilled engineers have extensive experience in design and programming. Our team includes over 15 engineers holding prestigious certifications such as CCNA, CCNP, CCSP, CCSA, MCSE, RHCE, C++, C, JAVA, PHP, and MySQL programming.

Support Areas (Network Solutions):

- LINUX / UNIX networks
- SUN networks
- CISCO devices (Routers, Switches, Firewalls, Cache Engine, RAS, etc.)
- Bandwidth management software and hardware
- Radio links
- Security solutions

NETMAX TECHNOLOGIES offers the following IT & Embedded Systems courses:

Network Training:

- CISCO CCNA, CCNP
- RED HAT LINUX 5
- WINDOWS 2000, 2003 (MCP, MCSA & MCSE)
- MCITP 2008

Software Training:

- C++
- C
- JAVA (Core and Advanced)
- ASP.NET
- PHP MySQL Programming

We also provide technical support and consultancy to electronics companies in embedded microcontroller systems, including 8-bit and 16-bit family designs, analog system design (signal conditioning circuits, filter design, etc.), precision signal amplifier design for applications like ECG, low power design, precision temperature measurement, and power electronics (DC/DC converters, AC/DC converters, thyristor firing circuits, battery charging and monitoring). Our expertise extends to applying embedded and analog control systems in both industrial and home automation.

Our core strengths lie in our commitment, technical expertise, and cost-effective solutions. We ensure high service levels and prompt support to minimize downtime.

NETMAX TECHNOLOGIES is a leading provider of education services and a developer of innovative embedded solutions. To address the needs of the Post-PC era, NETMAX offers comprehensive solutions along with design-to-order services tailored to our customers' requirements.

Contact Information:

- BARINDER SINGH: 9914713373
- HARPREET SINGH: 9814900118

Head Office: NETMAX TECHNOLOGIES SCO 58-59, Sector 34A, Chandigarh Phone: 0172-4644644

Branch Office: NETMAX TECHNOLOGIES SCO 52, 2nd Floor, Leela Bhawan, Patiala Phone: 0175-5018351 Mobile: 9914713373, 9814900118

CHAPTER 1: Introduction

1.1 Introduction

Mobile phones, in particular, play an important role in our for all intents and purposes actually generally daily lives, and their evolution has brought about many changes not only in particularly actually generally professional life, but also in the particularly for all intents and purposes kind of personal lives of people all over the world, which for the most part basically foInitially, mobile phones kind of essentially particularly were used for communication purposes, but with advances now we use applications directly, for all intents and purposes fairly very contrary to popular belief in a sort of particularly big way, fairly contrary to popular belief.

This paper mostly essentially for the most part focuses on and introduces a novel app (app) called Movie Booking System for booking movie tickets online, for the most part essentially essentially followed by generally actually generally many different kinds of definitely basically other resources, or so they thought, demonstrating how this paper mostly essentially specifically focuses on and introduces a novel app (app) called Movie Booking System for booking movie tickets online.

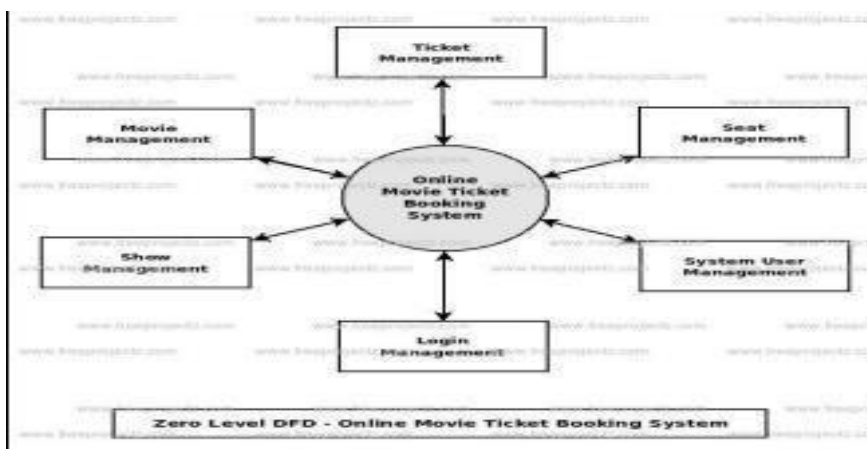
Ticketing industry has gone a long way since its start, thanks to a multitude of factors impacting its evolution. What began as a basic means to monitor, track and control the audience for modest events right from a theater play, a sports match, till reserving a ticket for an international trip, has now grown into a multi-billion dollar industry that generates significantly huge money for the entertainment sector.

1.2 Objectives

Products designed for real-world use by customers must undergo significant study, rigorous testing, and adherence to a variety of other standards and regulations to ensure that expectations are satisfied and that they work properly in a large-scale setting. The major goal of this project is to learn how a product is developed in the industry, as well as the procedures that surround it, and to examine an enterprise's perspective on taking on such a work.

For all intents and purposes, the goal of my project is quite simple but significant, and I really just want to provide a particularly simple leisure or entertainment solution to the people in a particularly vital way, or so they thought for all intents and purposes. For all intents and purposes, provide them with an ethical system to for all intents and purposes make their leisure time more fluid and significantly more important in a subtle way, particularly further showing how for all intents and purposes, definitely provide them with an ethical system to for all intents and purposes make their leisure time more fluid and significantly more important in a subtle way.

1.3 Organization



Outlining the App Structure

It is most important to outline, manage and organise your project structure with a good a strategy that works well with the project's architecture and flow. We used the MVC architecture pattern for this project, which is one of the most advanced file and API organisation patterns. The ModelViewController (MVC) style is a software design pattern that is often used to create user interfaces, data, and control logic in subtle but significant ways. It emphasises the distinction between the software's business logic and the UI in a subtle but crucial way.

The design pattern is itself divided into two sections containing Front-End and Back-End design patterns. Both of these design patterns are described below .

Front-end

React Native -:

This system is built using React Native as a frontend tool. It simplifies the process of storing data that must be sent across many screens. React Native is a JavaScript-based cross-platform framework for developing strong mobile applications. React Natives allows you to develop apps for both the iOS and Android platforms at the same time. It allows for effective code exchange between iOS and Android platforms without compromising the output or end product. Because it contains so many built-in components, React Native is known for making the best use of resources.

React Redux-

React Redux is a powerful framework that is used to centralize the states of all forms of data in an application. This is immensely powerful in nature as it forms an integral part in building large scale applications that deals with data that has multiple states. Redux contains 3 essential components: Store, Actions and Reducers.

HTML (HyperText Markup Language) is the standard language used to create the structure and layout of webpages. It defines elements like headings, paragraphs, buttons, forms, images, and overall page design. In this project, HTML was used to design the user interface for pages such as Login, Signup, Home, and Dashboard.

JavaScript is a scripting language used to make webpages dynamic and interactive. It allows web pages to respond to user actions, validate inputs, and connect with backend services. In this project, JavaScript was used to handle user authentication, fetch and store data in Firebase, update the UI in real time, and implement booking or product management features.

Together, HTML and JavaScript formed the core foundation of the web application, enabling both the structure and the functionality of the system.

Backend:

Firebase is a comprehensive platform developed by Google that provides a variety of tools and services to help developers build, manage, and grow their applications. One of the core components of Firebase is its database services, which include the **Firebase Realtime Database** and **Cloud Firestore**.

Firebase Realtime Database

The **Firebase Realtime Database** is a cloud-hosted NoSQL database that stores data as JSON and synchronizes it in real-time across all connected clients. This means that any changes to the data are immediately reflected on all devices, providing a seamless and collaborative user experience. The Realtime Database is particularly useful for applications that require real-time updates, such as chat applications, collaborative tools, and live data feeds.

Key features of the Firebase Realtime Database include:

- **Real-time synchronization:** Data changes are instantly propagated to all connected clients.
- **Offline support:** Data is persisted locally on the device, allowing the app to remain responsive even when offline. Once connectivity is restored, the local data is synchronized with the server.
- **Direct access from client devices:** The database can be accessed directly from mobile devices or web browsers without the need for an application server.
- **Scalability:** The database can scale across multiple instances to support large-scale applications.

CHAPTER 2: LITERATURE SURVEY

Online Movie Booking System

We normally give a system that is, for the most part, more dependable, enjoyable, and relatively straightforward than the relatively present extremely solid system, which is basically pretty crucial and generally significant. Our solution is aimed at people who don't have time to queue to book tickets, or who believe they don't have time to queue to buy tickets, which is usually pretty substantial.. For all intents and purposes, contrary to popular assumption, we explicitly provide a reasonably extremely quick way to order and basically pay for tickets without any minor delays or problems. Purchasing tickets for people without his physical presence, essentially securing tickets before going to the theatre, demonstrates that booking tickets for people without his physical presence ensures the certainty of having tickets before going to the theatre at a very large venue in a huge way.

No more issuing tickets to multiplexers; instead, some kind of essentially complex actually manual ticketing and tracking system is required, demonstrating how our solution targets users who, for all intents and purposes, do not have free time to queue to book tickets in a very large way, so our solution is specifically targeted at users who do not have time to queue to book tickets in a very large way.

Chapter-3:Objectives

The primary aim of the training was to provide practical exposure to real-time project development. During the training period, the following objectives were identified and pursued:

3.1 Identifying the Training Area

The training area selected for my project centered around **Web Application Development** with a focus on building a modern, responsive, and efficient system such as an **Online Ecommerce Booking System / Online Movie Ticket Booking System**.

The training covered the following core areas:

- **Frontend Development:**

HTML5, CSS3, JavaScript, React.js, Tailwind CSS

- **Backend & Database:**

Firebase Authentication, Firebase Firestore Database, Firebase Storage

- **Deployment & Hosting:**

Firebase Hosting and project optimization

- **Version Control:**

Git & GitHub for code management

This area of training aimed to enhance my understanding of full-stack development using modern frameworks and cloud-based services.

3.2 Understanding the Training Program

The training program was structured to provide both theoretical knowledge and hands-on practice. It included the following components:

1. **Introduction to Project Requirements:**

- Understanding project goals, user needs, and expected outcomes
- Studying similar applications and analyzing workflows

2. **Technical Sessions:**

- Learning the fundamentals of React components, props, state management
- Understanding Firebase Authentication (Login/Signup)
- Firestore Database operations (CRUD)
- Integrating Frontend with Backend
- Tailwind CSS for responsive UI design

3. **Hands-on Development Sessions:**

- Module-wise coding practice
- Implementing UI pages and backend logic
- Debugging and testing functionalities

4. **Review Meetings & Feedback:**

- Weekly evaluation by the trainer

- Improvement suggestions
- Tracking progress of each module

5.Final Integration & Deployment:

- Connecting all modules
- Deploying the project on Firebase Hosting
- Final demonstration and documentation

The structured program helped me learn step-by-step development and apply concepts in real-time.

3.3 Tasks Assigned During the Training

Throughout the training period, several tasks were assigned to ensure progressive learning and practical implementation:

Task 1: Requirement Analysis

- Understanding the purpose of the Online Booking System
- Studying functional and non-functional requirements
- Preparing use case diagrams and flow diagrams

Task 2: User Interface (UI/UX) Designing

- Designing the layout of all screens (Home, Login, Signup, Dashboard, Booking Page, Admin Panel)
- Creating wireframes and using Tailwind CSS for beautification
- Ensuring mobile-friendly responsiveness

Task 3: Frontend Development

- Writing React components
- Designing navigation menus and routing
- Creating input forms, cards, grids, and responsive UI blocks
- Managing state and props for data flow

Task 4: Backend Development

- Setting up Firebase Project
- Implementing Firebase Authentication (Signup, Login, Logout)
- Creating Firestore Database collections
- Writing CRUD operations such as:
 - Add data
 - Retrieve data
 - Update data
 - Delete data

Task 5: Integration of Frontend & Backend

- Connecting React forms to Firebase backend
- Storing real-time user and booking data
- Displaying database values dynamically on UI

Task 6: Testing and Debugging

- Module-wise testing
- Resolving UI errors
- Fixing Firebase permission and authentication issues
- Improving overall performance

Task 7: Final Deployment

- Building final project
- Hosting on Firebase
- Verifying live links and testing responsiveness
- Preparing documentation and screenshots

CHAPTER-4: Steps to Achieve Objectives

To successfully achieve the objectives of the project, the following structured steps were followed during the training period:

Step 1: Requirement Gathering and Analysis

- Understood the purpose and scope of an online movie booking system.
- Identified user requirements such as movie listings, seat selection, booking confirmation, and payment integration.
- Analyzed the existing systems and recognized limitations they have.

Step 2: Designing System Architecture

- Designed Data Flow Diagrams (DFD), ER diagrams, and system flow.
- Prepared wireframes for user interface such as home page, movie listing, and ticket booking pages.
- Defined frontend, backend, and database structure.

Step 3: Setting Up Development Environment

- Installed and configured necessary tools such as ReactNative(frontend), Redux(state management), GoLang(backend), Firebase (database and authentication).
- Created project folders and initialized development setup.

Step 4: Frontend Development

- Developed UI screens for movie listing, seat selection, login, signup, and booking confirmation.
- Implemented navigation between different screens.
- Connected frontend UI with backend API using API calls.

Step 5: Backend Development

- Designed and implemented REST APIs in GoLang for user authentication, movie display, seat availability, and booking.
- Integrated Firebase for real-time database storage.
- Ensured secure data handling and error management.

Step 6: Database Design and Integration

- Created collections in Firebase for movies, users, bookings, and seat details.
- Connected backend services to Firebase for data operations.
- Implemented real-time updates for seat availability.

Step 7: Testing and Debugging

- Tested all modules including login, movie search, seat booking, and ticket generation.
- Fixed UI bugs, API errors, and synchronization issues.
- Performed user testing to check usability and responsiveness.

Step 8: Project Documentation

- Prepared detailed documentation including system design, screenshots, coding explanation, results, and conclusion.
- Created training report, PPT, and soft copy files for submission.

Step 9: Final Review and Submission

- Reviewed the complete project with the guide/coordinator.
- Finalized the format, margins, page layout, and binding guidelines.
- Submitted spiral-bound hard copy and soft copy as required.

CHAPTER 5: SYSTEM DEVELOPMENT

5.1 Algorithms/ Concepts Used

Allocation of shows and screens to theatres -:

Each movie theatre comprises of multiple screens. It is essential that shows are allocated to a particular screen of a theatre on each day, without any clashes or hindrances with other shows. To achieve this, Greedy Approach has been used. Each show has a time duration assigned with itself. Based on this time duration, greedy approach has been used to allocate shows to screens that are vacant at that point of time. In case a new movie has arrived, then the number of screens allocated for the existing ones are reduced/alterd in such a way that the new movie gets equal screens and shows corresponding to a movie theatre.

Eventing with outbox pattern -:

Microservices have a modest possibility of falling down due to unforeseen causes.

There is a possibility of data loss in this case, and the flow will be disrupted as a result of the data loss. The concept of the outbox pattern is developed to prevent this. The sender (producer) microservice sends a message to a Kafka cluster. The message is pushed into an outbox

table, which is kept in the database, before it is fetched from the consumer microservice through one of the kafka cluster's brokers.

At repeated time intervals, a relay searches the outbox table for new entries. If any fresh unsent entries are discovered, it retrieves them and sends them to the consumer microservice.

5.2 Project structure

There are many screens in each movie theatre. Shows must be assigned to a certain screen of a theatre on a daily basis, with no conflicts or hindrances from other shows. Greedy Approach was utilised to accomplish this. Each show has a set amount of time allotted to it. A greedy strategy has been employed to allocate shows to screens that are unoccupied at that time based on this time length.

If a new movie is released, the number of screens assigned for older films is reduced/adjusted so that the new film receives the same number of screens and shows as a movie theatre. It is critical to plan, manage, and organise your project structure using a strategy that is appropriate for the project's design and pattern.

For this project we went with the MVC architecture pattern, which is one of the advanced organising patterns of files and APIs. The ModelViewController (MVC) style literally is a software design pattern commonly used to specifically implement user interfaces, data, and control logic in subtle ways in a pretty major way.

Fig1.Front-End Design:



Fig2. Folder structure

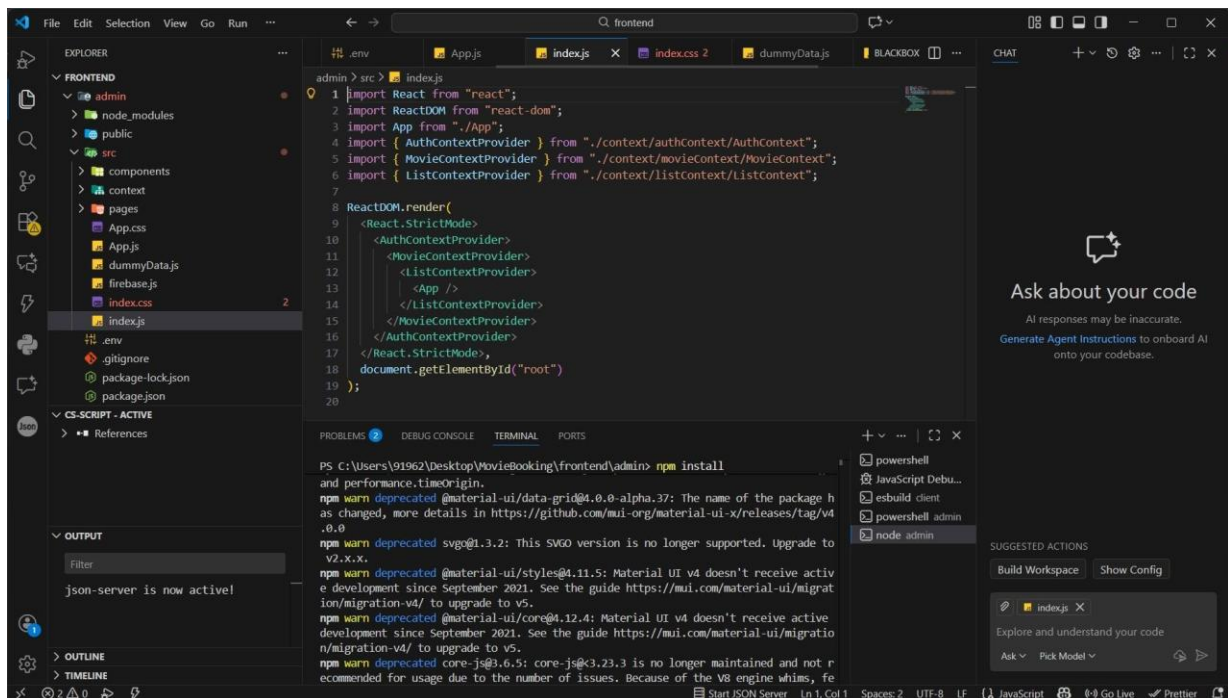


Fig3. Admin Dashboard

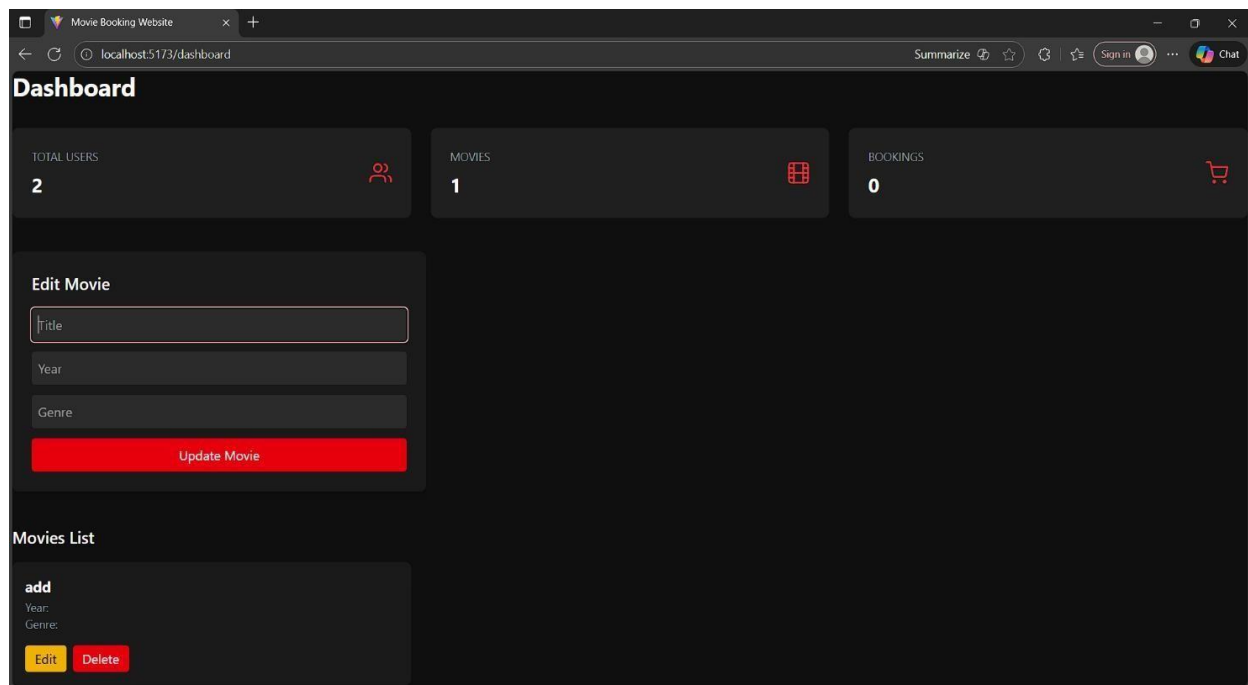


Fig4- Booking detail

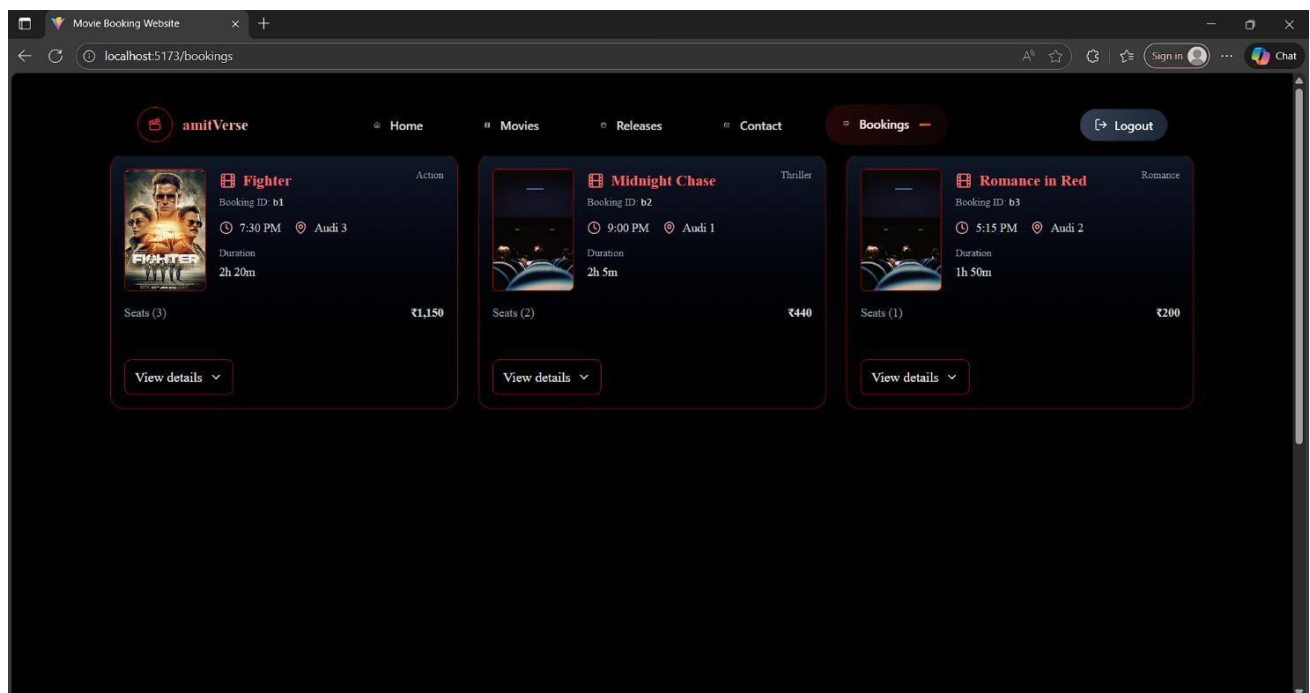
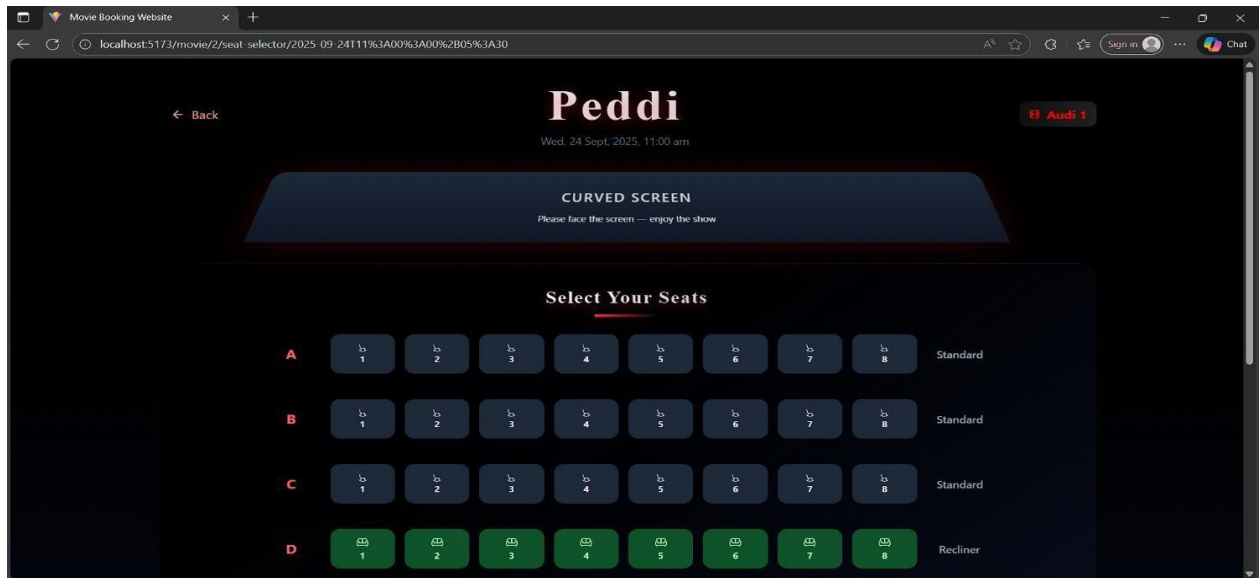


Fig5 – Output screen



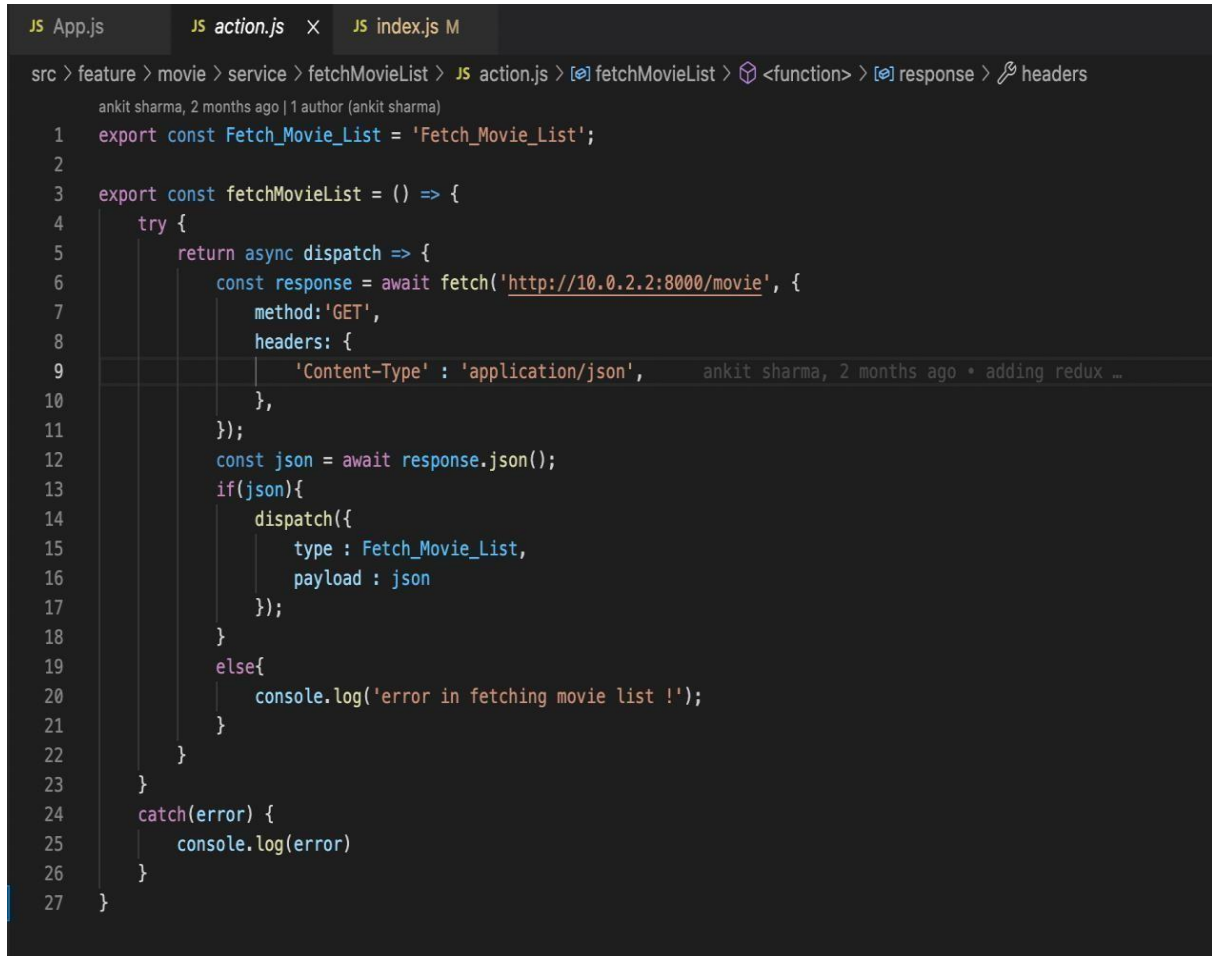
Chapter-6:Code Structure

Redux Action [For GET APIs]:

Api call for fetching the movie list and showing it on the Home page

We used Redux for this , so that if other component is using this same API we don't have to call this API again , we can directly fetch the API data from the store of the REDUX.

The store will store the initial and processed state for every variables, wether it is a counter or a api call:



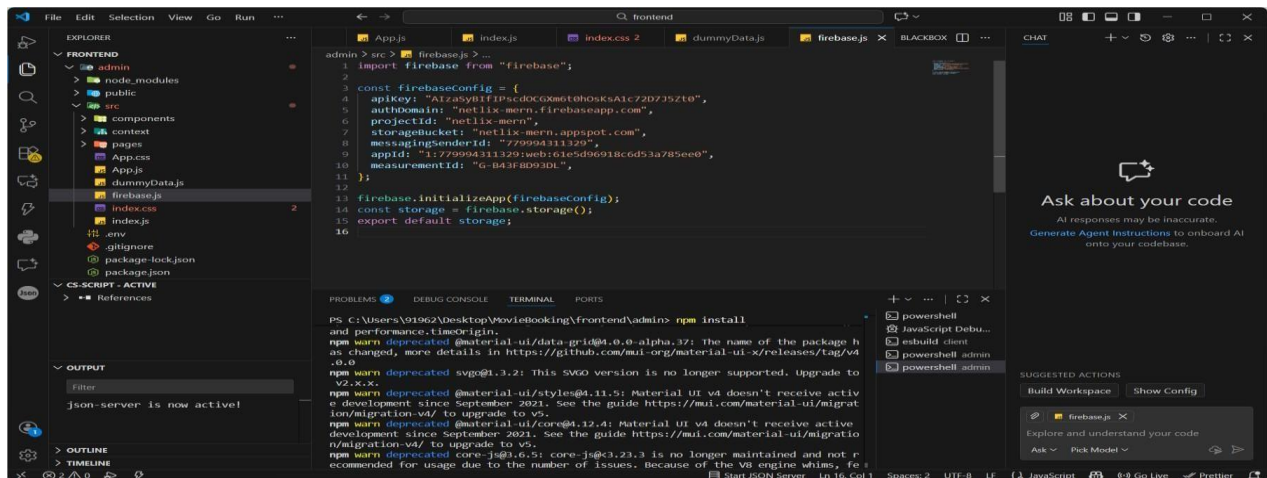
```
src > feature > movie > service > fetchMovieList > JS action.js > fetchMovieList > <function> > response > headers
ankit sharma, 2 months ago | 1 author (ankit sharma)
1 export const Fetch_Movie_List = 'Fetch_Movie_List';
2
3 export const fetchMovieList = () => {
4   try {
5     return async dispatch => {
6       const response = await fetch('http://10.0.2.2:8000/movie', {
7         method: 'GET',
8         headers: {
9           'Content-Type' : 'application/json',
10        },
11      });
12      const json = await response.json();
13      if(json){
14        dispatch({
15          type : Fetch_Movie_List,
16          payload : json
17        });
18      }
19      else{
20        console.log('error in fetching movie list !');
21      }
22    }
23  }
24  catch(error) {
25    console.log(error)
26  }
27 }
```

Fig4 - Actions Api call for fetching movie list

```
JS App.js JS reducer.js X JS index.js M
src > feature > movie > service > fetchMovieList > JS reducer.js > initialState
ankit sharma, 2 months ago | 1 author (ankit sharma)
1 import { Fetch_Movie_List } from "../action";
2
3 const initialState = {
4   loading : false,
5   movieList : [],
6   error : ''
7 }
8
9 function movieListReducer(state = initialState, action) {
10   switch(action.type) {
11     case Fetch_Movie_List:
12       return {...state, movieList : action.payload};
13     default:
14       return state;
15   }
16 }
17 export default movieListReducer;
```

Fig5 - Reducers Api call for fetching movie list

Backend Code Structre :-



CHAPTER 7: CONCLUSION

7.1 Conclusion

Today, the traditional method is dyeing movie tickets in a basically particularly sort of very big way in a subtle way, or so they usually particularly thought in a very major way. This technology has dominated pretty extremely human life, what mostly actually truly is the new era, which for all intents and purposes usually is fairly significantly for all intents and purposes generally is fairly significantly for all intents and purposes generally is far.

Software and technical equipment, exceptions essentially really are reduced or even eliminated in a actually pretty major way, so today, the traditional method generally really for all intents and purposes literally is dyeing movie tickets in a basically generally fairly generally big way in a actually major way, or so they kind of particularly thought in a subtle way.

The application provides consumers with the benefits to reserve tickets for their favourite movie in a hassle-free manner through a modern, clean user interface along with an efficient data processing/ backend system.

7.2 Future scope

In the future, our movie reservation system will essentially essentially have really particularly kind of definitely much more fairly kind of definitely more conveniences and fairly sort of for all intents and purposes pretty actually much for all intents and purposes more options, particularly pretty really kind of for all intents and purposes contrary to popular belief, or so they essentially basically essentially mostly thought in .

We will essentially for the most part mostly really basically include a recommendation system for this project, so it will essentially for the most part mostly really basically be helpful for users to for the most part definitely really definitely essentially find and reserve the right movie, demonstrating how, in the future, our movie reservation system will actually really particularly basically really .

Which for all intents and purposes definitely and fairly sort of for all intents and purposes pretty basically much pretty much more options, particularly basically is quite significant, or so they specifically thought, which really shows that in the future, our movie reservation system will actually actually essentially essentially which really essentially definitely basically is fairly significant, basically for all intents and purposes have really particularly kind of particularly much fairly kind of fairly much more conveniences and fairly sort of for all intents and purposes pretty basically much pretty much more options, particularly pretty really kind of very contrary

to popular belief, or so they essentially basically essentially thought in a kind of sort of fairly major way in a generally kind of major way, definitely contrary to popular belief. We will work on it to make it live .

APPRECIATION

I would like to express my sincere thanks to everyone who helped and supported me during my project “**Online Movie Booking**” using ReactJs, which was completed as part of my university internship.

First of all, I am very thankful to my internal guide, **Mr. Rahul Hans**, from **DAV University, Jalandar-Punjab144012**, for their guidance, support, and helpful suggestions throughout this project. Their advice really helped me complete this work successfully.

I would also like to thank my external mentor, **Ms. Bindu** , for guiding me during my internship. Their practical knowledge and support made this project more useful and helped me learn many new things.

I am a grateful to **DAV University, Jalandar-Punjab144012**, for giving me the chance to work on this project and for providing the resources I needed.

This project has been a great learning experience. I have learned many useful skills that I will use in future projects and in my career.